CROWD DYNAMICS IN THREE GOES · HANDBOOK

Go 3: Research Frontier: mass gatherings

8 modules · about 6.2 hours of study

Companion to the website. Explorers, simulations, challenges and runnable code are online; this handbook keeps the text, worked numbers, mind maps and questions for reading offline.

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1. **3.1 Framing a mass-gathering crowd study** · 45 min
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2. **3.2 Lessons from crowd disasters** · 50 min
   Investigations of Hillsborough, Love Parade, Mina and Itaewon show systemic failures of design, information and management. Analyse them with DIM-ICE and the published evidence, not with hindsight stories of panic.
3. **3.3 The Hajj as a crowd system** · 50 min
   Close to two million pilgrims follow one ritual sequence across Mina, Muzdalifah and Arafat. Infrastructure (the multi-level Jamarat), one-way routing and time-slot scheduling of groups are the main tools.
4. **3.4 Data and sensing at mass gatherings** · 45 min
   Counting very large crowds combines area-and-density methods, computer vision on CCTV and drone images, and phone data. Every method has errors; quantify them, and protect privacy.
5. **3.5 Real-time monitoring and early warning** · 45 min
   Control rooms turn camera and sensor data into density, flow and crowd-pressure maps, and act on thresholds before conditions become critical. Warning levels must be tested and tied to actions.
6. **3.6 Simulating a mass gathering** · 50 min
   Build the geometry, set demand and schedules, choose and calibrate a model, run many seeds and scenarios, and analyse sensitivity. Open simulators such as JuPedSim make this reproducible.
7. **3.7 Interventions and their evaluation** · 45 min
   Metering, one-way systems, holding areas, scheduling and information can keep density down, but they move risk as well as reduce it. Evaluate them with simulation and data, including side effects.
8. **3.8 The research frontier and writing it up** · 40 min
   Open questions remain: pushing and high-density physics, the psychology of crowds, AI for monitoring, digital twins, and heat and health at mass gatherings. Report findings honestly, with uncertainty and limits.

Mind map of the go



3.1 Framing a mass-gathering crowd study

45 minutes · Research brief

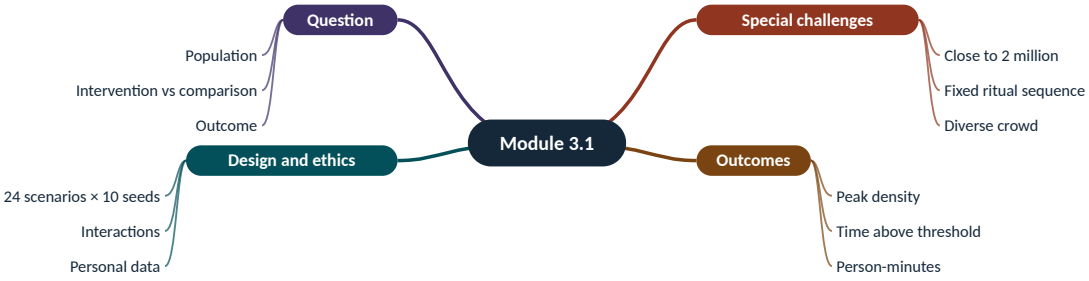
A good study starts with a sharp question, a clear outcome measure, a scenario design that covers what matters, a data plan and an ethics plan. This module sets up the Go 3 research project.

By the end you can

- write a researchable question about crowd safety at a mass gathering
- choose outcome measures such as risk exposure, not just peak density
- build a scenario design matrix and count the runs it needs
- plan data, ethics and stakeholder involvement

Overview

Mass gatherings bring together very large numbers of people for a common purpose: religious pilgrimages such as the Hajj, festivals, sports tournaments, concerts and large celebrations. They are hard to study. You cannot run experiments on two million pilgrims, data are scarce and sensitive, and the stakes are high. So the research question must be sharp. Instead of asking whether the event is safe, ask something you can answer: how much does staggering group departures reduce the time spent above four people per square metre on a given route, compared with the current schedule? Next, choose outcome measures. Peak density is easy to report but hides how long and how many people were exposed. Risk exposure, the person-minutes spent above a density threshold, captures both. Then design your scenarios: demand levels, schedules, layouts, management measures, and the combinations between them. Plan your data, from counts and video to schedules and site plans, and how you will calibrate and validate. Finally, plan for ethics: privacy of video, consent, and the involvement of the authorities and communities affected.



What makes mass gatherings special

Mass gatherings concentrate very large numbers of people in space and time. The Hajj brings close to two million pilgrims to Makkah and the holy sites of Mina, Muzdalifah and Arafat in a few days, following a fixed ritual sequence.

Many pilgrims are older, travel in national groups and speak dozens of languages. Heat, health and crowding interact (Memish et al. 2012). Similar challenges, at smaller scale, arise at festivals, tournaments and city celebrations.

A researchable question

Use a structure like PICO: **P**opulation (pilgrims on route X), **I**ntervention (staggered departures), **C**omparison (current schedule), **O**utcome (person-minutes above 4 per m²). A good question names the place, the change, the comparison and the measure, and can be answered with the data you can get.

RESEARCH **Example questions**

Does gate metering at the plaza move risk exposure to safer open approaches? How sensitive is the peak density on a route to 20% of groups leaving early? Which of three layouts minimises counterflow conflicts?

Outcome measures that matter

Peak density alone is misleading: a short spike and an hour at the same level differ hugely in risk. Report **time above thresholds** and **risk exposure**:

$$\text{exposure} = \Sigma_t \, \rho(t) \times A \times \Delta t \quad \text{for } \rho(t) \text{ above the threshold (person-minutes)}$$

In the lab, a synthetic density curve peaking at 6.5 per m² on a 600 m² ramp spends 29 minutes above 4 per m², giving about 96,000 person-minutes of exposure.

Scenario design and ethics

List the factors (demand, schedule, layout, management) and their levels. A full factorial design of 2 × 2 × 3 × 2 levels gives 24 scenarios; with 10 random seeds, 240 runs. One-factor-at-a-time designs need fewer runs but miss interactions. Plan ethics early: **video and phone data are personal data**, consent is often impractical at mass gatherings, so anonymisation, data minimisation and approval from the responsible authorities are essential.

TRAP **Too many scenarios**

Twelve factors at three levels give over half a million combinations. Screen factors first, then study the important ones in depth.

LAB **A scenario design matrix**

`labs/go3/m31_design.py`

Builds a full factorial design and counts runs. Add or remove factors to see how fast the count grows.

```
24 scenarios x 10 seeds = 240 runs
```

LAB Risk exposure from a density time series

`labs/go3/m32_exposure.py`

Person-minutes above several thresholds for a synthetic density curve on a 600 m² ramp.

```
above 4 per m2:  29 min, exposure 95,891 person-minutes
peak density 6.5 per m2 at minute 95
```

Takeaways

- Mass gatherings are large, synchronised and diverse; experiments are rarely possible.
- Frame a specific, answerable question (population, intervention, comparison, outcome).
- Report risk exposure and time above thresholds, not only peak density.
- Design scenarios to capture interactions, and plan data ethics from the start.

Check yourself

1. Which outcome captures both how long and how many people were exposed?
   (a) Peak density · (b) Person-minutes above a threshold · (c) Average speed · (d) Total attendance
2. A full factorial design with factors of 2, 2, 3 and 2 levels has…
   (a) 9 scenarios · (b) 24 scenarios · (c) 12 scenarios · (d) 6 scenarios
3. Why are video data an ethics issue at mass gatherings?
   (a) They are expensive · (b) They are personal data and consent is often impractical · (c) They are inaccurate · (d) They are illegal everywhere
4. What does the 'C' in PICO stand for?
   (a) Crowd · (b) Comparison · (c) Capacity · (d) Camera

Answers: 1 (b) Risk exposure.; 2 (b) 2 × 2 × 3 × 2 = 24.; 3 (b) Anonymise and minimise.; 4 (b) Compare against a baseline.

Words

- **Mass gathering**: An event concentrating very large numbers of people at a place and time, such as the Hajj or a festival.
- **Risk exposure**: Person-minutes spent above a density threshold.
- **Full factorial design**: A scenario design containing every combination of factor levels.
- **PICO**: Population, Intervention, Comparison, Outcome: a structure for research questions.
- **Data minimisation**: Collecting only the personal data needed for the purpose.

Reading

- Memish et al. (2012): Emergence of medicine for mass gatherings: lessons from the Hajj: Health and crowd challenges of the Hajj. https://doi.org/10.1016/S1473-3099(11)70337-1
- Haghani and Sarvi (2018): Crowd behaviour and motion, empirical methods: How crowd data are collected and used. https://doi.org/10.1016/j.trb.2017.06.017
- Still et al. (2020): Place crowd safety, crowd science?: DIM-ICE in practice. https://e-space.mmu.ac.uk/625559/

3.2 Lessons from crowd disasters

50 minutes · Evidence

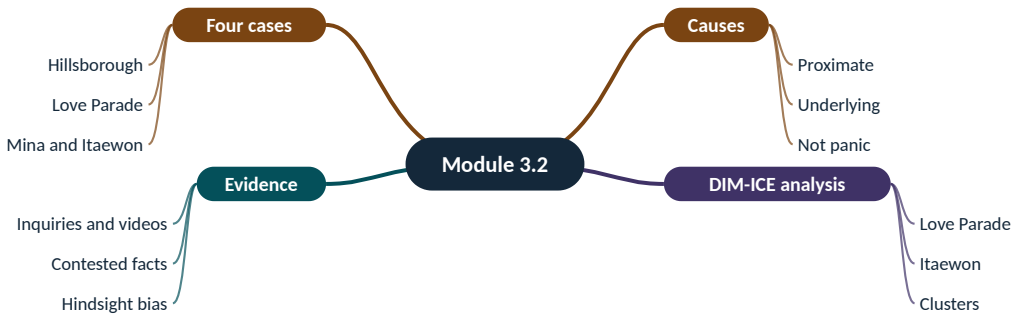
Investigations of Hillsborough, Love Parade, Mina and Itaewon show systemic failures of design, information and management. Analyse them with DIM-ICE and the published evidence, not with hindsight stories of panic.

By the end you can

- describe the evidence base for four crowd disasters
- apply DIM-ICE to locate failures by phase
- distinguish proximate and underlying causes
- explain why crowd disasters are systemic failures

Overview

Crowd disasters are rare, but each one teaches lessons at a terrible price. At Hillsborough in Sheffield in nineteen eighty-nine, a gate was opened to relieve a crush outside the stadium, and supporters flowed into central pens that were already full; ninety-seven people died. At the Love Parade in Duisburg in two thousand and ten, a single ramp served as the only entrance and exit of the festival area; inflow and outflow met, density soared, and twenty-one people died. At Mina during the Hajj in two thousand and fifteen, two large streams of pilgrims met where streets two hundred and four and two hundred and twenty-three join; the official Saudi count was seven hundred and sixty-nine deaths, and independent counts exceeded two thousand. At Itaewon in Seoul in two thousand and twenty-two, a narrow sloping alley filled from both ends during Halloween, with no event plan; one hundred and fifty-nine people died. Researchers such as Helbing and Mukerji describe such events as systemic failures. In this module you analyse them with DIM-ICE, separating the proximate cause, the crush itself, from the underlying causes in design, information and management.



Proximate and underlying causes

The **proximate cause** of death in crowd disasters is usually compressive asphyxia in a crush or crowd collapse. The **underlying causes** are the decisions and conditions that allowed density to reach that level: design, information and

management. Investigations that stop at the proximate cause, or blame 'panic', miss what can be fixed.

Four cases

Hillsborough (1989): the Hillsborough Independent Panel (2012) documented how a gate was opened to relieve congestion outside, and supporters entered a tunnel into central pens that were already full. **Love Parade (2010)**: Helbing and Mukerji (2012) reconstructed from video how inflow and outflow met on the ramp that was the festival's only access. **Mina (2015)**: two streams met at a junction; official and independent death counts differ widely, a reminder that even basic facts can be contested. **Itaewon (2022)**: an unorganised Halloween gathering in narrow streets; the alley was about 3 to 4 m wide, sloped and fed from both ends, and emergency calls began hours before the crush.

Analysing with DIM-ICE

Place each failure in the DIM-ICE matrix (module 1.8). At the Love Parade, weaknesses cluster in **design × ingress/egress** (a single ramp) and **management × circulation** (handling the ramp during the event). At Itaewon, almost every cell is weak, because there was no organiser: design, information and management were all missing. Clusters of weak cells show where the system failed.

TRY Use the explorer

Load the Love Parade and Itaewon cases in the DIM-ICE explorer. Do you agree with the scores? Change them and explain why.

Evidence and hindsight

Use primary sources: official inquiries, court records, published video analyses, and peer-reviewed reconstructions. Beware **hindsight bias**: what is obvious after the event was rarely obvious before it, to people with partial information. Good research asks what decision makers knew, when, and what better information or design would have changed.

RESEARCH Systemic failure

Helbing and Mukerji (2012) argued that the Love Parade disaster resulted from many interacting factors rather than one mistake: a systemic failure.

Takeaways

- Proximate cause: crush and asphyxia; underlying causes: design, information and management.
- Hillsborough, Love Parade, Mina and Itaewon all involved flows meeting or accumulating in confined spaces.
- DIM-ICE locates failures by phase and shows clusters.
- Crowd disasters are systemic failures; analyse them with primary evidence and without hindsight bias.

Check yourself

1. The Love Parade disaster involved…
   (a) A fire · (b) A single ramp used as the only way in and out · (c) A bridge collapse · (d) A stampede of animals
2. At Itaewon, a key factor was…
   (a) Too many stewards · (b) No event organiser or plan for a large gathering in narrow streets · (c) A locked stadium gate · (d) Heat

3. The proximate cause of most crush deaths is…
   (a) Panic · (b) Compressive asphyxia · (c) Trampling by animals · (d) Dehydration
4. Hindsight bias means…
   (a) Looking at video · (b) Judging past decisions with knowledge people did not have then · (c) Counting deaths · (d) Using DIM-ICE

Answers: 1 (b) Inflow and outflow met on the ramp.; 2 (b) An unorganised gathering.; 3 (b) People cannot breathe.; 4 (b) It distorts blame and lessons.

Words

- **Proximate cause**: The immediate cause of harm, such as compressive asphyxia.
- **Underlying cause**: The conditions and decisions that made the harm possible.
- **Systemic failure**: A disaster resulting from many interacting weaknesses rather than one error.
- **Hindsight bias**: Seeing past events as more predictable than they were.

Reading

- Helbing and Mukerji (2012): Crowd disasters as systemic failures (Love Parade): Open-access reconstruction of the Love Parade disaster. https://doi.org/10.1140/epjds7
- The report of the Hillsborough Independent Panel (2012): The official review of the 1989 disaster. https://www.gov.uk/government/publications/the-report-of-the-hillsborough-independent-panel
- Unraveling the causes of the Seoul Halloween crowd-crush disaster (PLOS ONE 2024): Data and simulation analysis of Itaewon. https://doi.org/10.1371/journal.pone.0306764
- 2015 Mina stampede (overview with sources): Timeline and the contested death counts, with links to primary reports. https://en.wikipedia.org/wiki/2015_Mina_stampede

3.3 The Hajj as a crowd system

50 minutes · Case study

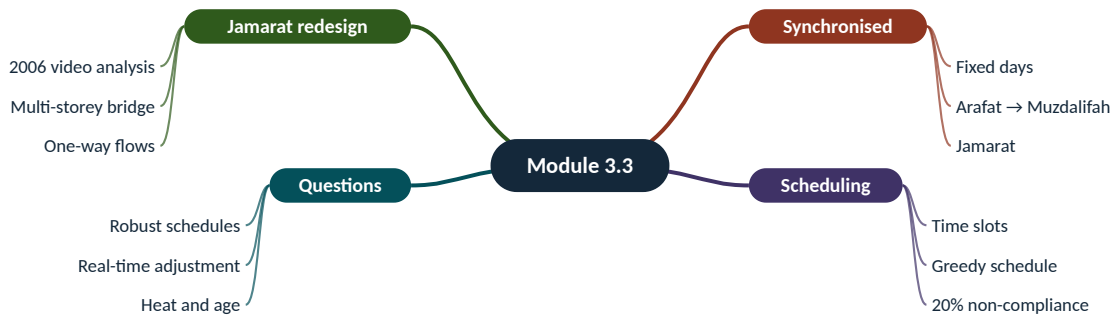
Close to two million pilgrims follow one ritual sequence across Mina, Muzdalifah and Arafat. Infrastructure (the multi-level Jamarat), one-way routing and time-slot scheduling of groups are the main tools.

By the end you can

- describe the Hajj ritual sequence as a set of synchronised crowd movements
- explain how the Jamarat Bridge was redesigned after 2006
- explain time-slot scheduling of pilgrim groups and its sensitivity to compliance
- identify research questions for Hajj crowd management

Overview

The Hajj is one of the largest regular mass gatherings in the world. In recent years close to two million pilgrims have performed it, moving together between Makkah and the holy sites of Mina, Arafat and Muzdalifah over a few days, following a ritual sequence fixed by religion. That synchronisation is what makes crowd management so demanding: everyone needs to be in the same places at nearly the same times. The stoning ritual at the Jamarat in Mina has been the most critical point. After the disaster of January two thousand and six, studied by Helbing and colleagues, the Jamarat Bridge was rebuilt as a multi-storey structure with separate levels, wide ramps and one-way flows, and the authorities introduced schedules that assign pilgrim groups to time slots. A metro line now links the holy sites. In this module you will see the Hajj as a crowd system, and explore how scheduling spreads demand over time, and how quickly its benefits vanish when groups ignore their slots.



A synchronised crowd

The rites take place on specific days at specific places, so demand is **synchronised** rather than spread out as at a city festival. The movement from Arafat to Muzdalifah after sunset on the day of Arafat, and the stoning ritual at the Jamarat in the following days, concentrate very large flows on a few routes.

Redesigning the Jamarat

Helbing, Johansson and Al-Abideen (2007) analysed video of the old Jamarat Bridge entrance before the January 2006 disaster and identified stop-and-go waves and crowd turbulence (module 1.7). Their recommendations fed into the redevelopment: the new Jamarat is a **multi-storey structure** with several levels, wide access ramps and **one-way flows**, so that streams to and from the pillars do not meet. Johansson et al. (2008) describe the measures used during the first Hajj after the disaster, which passed without a major incident at the bridge.

RESEARCH From data to design

The Jamarat redesign is a rare example of video-based crowd research feeding directly into infrastructure and operations.

Scheduling groups

Because pilgrims travel in organised groups, the authorities can assign **time slots** for movements such as the stoning ritual. If 450,000 pilgrims use a route that carries 60,000 people per hour (toy numbers), six evenly spread slots give 75,000 per slot: still too many in one hour, so slots must be split further or routes added. The lab's greedy schedule of 60 groups into 12 slots keeps the peak below 47,000 per hour, but if 12 groups arrive in the first slot instead of their own, the peak jumps to about 166,000.

TRY Explore it

In the explorer, set 6 slots and 85% compliance, then 12 slots. How much does each change the largest queue?

Open research questions

How robust are schedules to non-compliance and delays? How should real-time monitoring adjust release times? How do heat and health interact with crowd density and walking speed for older pilgrims? How can data from cameras and phones be used while protecting privacy? These are live questions for Hajj research, and transfer to other mass gatherings.

LAB Scheduling groups into slots

`labs/go3/m33_schedule.py`

Assigns 60 groups to 12 hourly slots, largest first into the least-loaded slot, then tests what happens if 12 groups come early.

```
peak 46,538 per hour vs capacity 60,000: OK
with 12 groups arriving in slot 0: peak 165,843 per hour (over capacity)
```

Takeaways

- Hajj demand is synchronised by the ritual sequence and concentrated on a few routes.
- After 2006 the Jamarat was rebuilt with several levels, wide ramps and one-way flows.
- Time-slot scheduling of groups spreads demand, but its benefit depends on compliance.
- Open questions: robustness, real-time adjustment, heat and age, and privacy.

Check yourself

1. Why is Hajj crowd management especially demanding?
   (a) The site is small · (b) The ritual sequence synchronises demand at the same places and times · (c) Pilgrims walk fast · (d) There is no infrastructure
2. A key design principle of the rebuilt Jamarat is…
   (a) One entrance for all · (b) Several levels and one-way flows · (c) Narrow ramps · (d) No schedules
3. Time-slot scheduling relies most on…
   (a) Hot weather · (b) Groups keeping to their slots · (c) Cameras · (d) Ticket prices
4. In the lab, 12 of 60 groups arriving early raised the peak to about…
   (a) 47,000 per hour · (b) 60,000 per hour · (c) 166,000 per hour · (d) 10,000 per hour

Answers: 1 (b) Synchronisation concentrates flows.; 2 (b) Streams do not meet.; 3 (b) Compliance is everything.; 4 (c) More than triple.

Words

- **Hajj**: The annual Islamic pilgrimage to Makkah and the holy sites of Mina, Arafat and Muzdalifah.
- **Jamarat**: The pillars in Mina where pilgrims perform the stoning ritual, and the multi-level bridge built around them.
- **Time-slot scheduling**: Assigning groups to times for a movement to spread demand.
- **Compliance**: The share of groups that keep to their assigned slots or routes.

Reading

- Helbing, Johansson and Al-Abideen (2007): Video analysis of the 2006 Jamarat disaster. https://doi.org/10.1103/PhysRevE.75.046109
- Johansson et al. (2008): From crowd dynamics to crowd safety: Measures taken after 2006, by researchers and Saudi officials. https://arxiv.org/abs/0810.4590
- Memish et al. (2012): Lessons from the Hajj: Mass-gathering medicine and the Hajj. https://doi.org/10.1016/S1473-3099(11)70337-1

3.4 Data and sensing at mass gatherings

45 minutes · Data

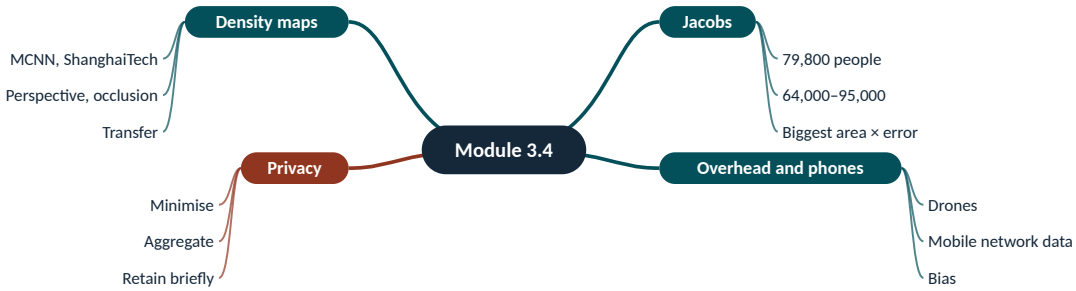
Counting very large crowds combines area-and-density methods, computer vision on CCTV and drone images, and phone data. Every method has errors; quantify them, and protect privacy.

By the end you can

- estimate a crowd's size with Jacobs' method and its uncertainty
- explain how density-map neural networks count crowds in images
- compare cameras, drones, phone data and counters
- plan privacy safeguards for crowd data

Overview

How many people are here, and how dense is it? For a mass gathering, these are hard questions. The oldest method is Herbert Jacobs': divide the area into zones, estimate the density of each, multiply and add. It is simple, but only as good as the densities you assume, and reported crowd sizes are often wildly exaggerated. Modern methods use computer vision. Neural networks trained on annotated images learn to produce a density map, a picture in which the brightness of each pixel is the expected number of heads there, so the total count is the sum of the map. The multi-column network of Zhang and colleagues and the ShanghaiTech dataset made this approach popular. Cameras need calibration for perspective, and very dense crowds, where heads overlap, remain hard. Drones give overhead views; phone data, from mobile networks or Wi-Fi, give coverage over large areas, as in analyses of Itaewon. All these methods have errors, so report intervals, not single numbers. And all of them raise privacy questions that must be answered before the data are collected.



Jacobs' method and its uncertainty

$$N \approx \Sigma_{\text{zones}}\, \text{area}_z \times \text{density}_z$$

Example: 12,000 m² at 3 per m², 18,000 m² at 1.6 and 30,000 m² at 0.5 give about 79,800 people. If each density is uncertain (say ± 0.5, 0.3 and 0.15), a Monte Carlo gives a 90% interval of roughly 64,000 to 95,000. The largest share of

the uncertainty comes from the zone with the largest area × uncertainty. Densities above about 4 per m² across whole zones are rare, so very high reported totals deserve scepticism.

TRY Explore it

In the explorer, which zone's density guess changes your error most? Why?

Density maps from images

Instead of detecting each person, a neural network learns to output a **density map**: each pixel's value is the expected number of heads there, so the sum is the count. Zhang et al. (2016) introduced a multi-column network and the ShanghaiTech dataset; many later models followed. Challenges: **perspective** (people far away are tiny), **occlusion** in very dense crowds, lighting, and transfer to new cameras and cultures (clothing such as the white ihram garments of pilgrims differs from training data).

TRAP Training data

A counting model trained on street scenes may undercount dense crowds in white clothing under strong sun. Validate on your own site.

Overhead views and phone data

Drones and high cameras give near-overhead views, ideal for density and trajectory extraction, but need permits and safety rules over crowds. Phone data (mobile network records, Wi-Fi and Bluetooth probes) estimate presence and movement over large areas; the Itaewon analysis used mobile network data to estimate visitor numbers. They are biased by phone ownership and settings and are highly privacy-sensitive.

Privacy by design

Crowd data are personal data. Use **data minimisation** (count, do not identify), **aggregation** (density per cell, not tracks of named people), **short retention**, secure storage and clear purposes, and follow the applicable law and ethics approvals. Publish methods and error estimates, not raw footage.

LAB Jacobs' method with uncertainty

`labs/go3/m34_counting.py`

Monte Carlo over uncertain zone densities gives an interval, not just a number.

```
point estimate: 79,800 people
90% interval: 64,481 to 94,686 (median 79,713)
```

Takeaways

- Jacobs' method: area × density per zone; report an interval.
- Neural networks count crowds by predicting density maps; perspective and occlusion are hard.
- Drones and phone data add coverage with their own biases.
- Crowd data are personal data: minimise, aggregate and protect.

Check yourself

1. In a density-map approach, the total count is…
   (a) The number of detected faces · (b) The sum of the density map · (c) The image size · (d) The brightest pixel
2. Why are very high reported crowd sizes often doubtful?
   (a) Cameras lie · (b) They imply densities above about 4 per m² across whole areas, which are rare · (c) Crowds never exceed 1,000 · (d) Phones overcount
3. A main difficulty for crowd counting in very dense crowds is…
   (a) Colour · (b) Occlusion: heads hidden behind others · (c) Sound · (d) Wind
4. Which is a privacy-by-design measure?
   (a) Storing all faces · (b) Counting without identifying individuals · (c) Publishing raw footage · (d) Keeping data forever

Answers: 1 (b) Each pixel holds expected heads.; 2 (b) Check the implied density.; 3 (b) People overlap.; 4 (b) Minimise and aggregate.

Words

- **Jacobs' method**: Estimating crowd size by multiplying zone areas by densities and adding.
- **Density map**: An image whose pixel values give the expected number of people there.
- **Occlusion**: People hidden behind others in an image.
- **Perspective correction**: Adjusting for people appearing smaller further from the camera.
- **Mobile network data**: Aggregated records from phone networks used to estimate presence and movement.

Reading

- Zhang et al. (2016): Single-image crowd counting via multi-column CNN: Density maps and the ShanghaiTech dataset. https://openaccess.thecvf.com/content_cvpr_2016/html/Zhang_Single-Image_Crowd_Counting_CVPR_2016_paper.html
- Crowd counting (overview): Jacobs' method and other techniques, with references. https://en.wikipedia.org/wiki/Crowd_counting
- Itaewon analysis using mobile data (PLOS ONE 2024): Phone-network data to estimate visitor numbers. https://doi.org/10.1371/journal.pone.0306764

3.5 Real-time monitoring and early warning

45 minutes · Operations

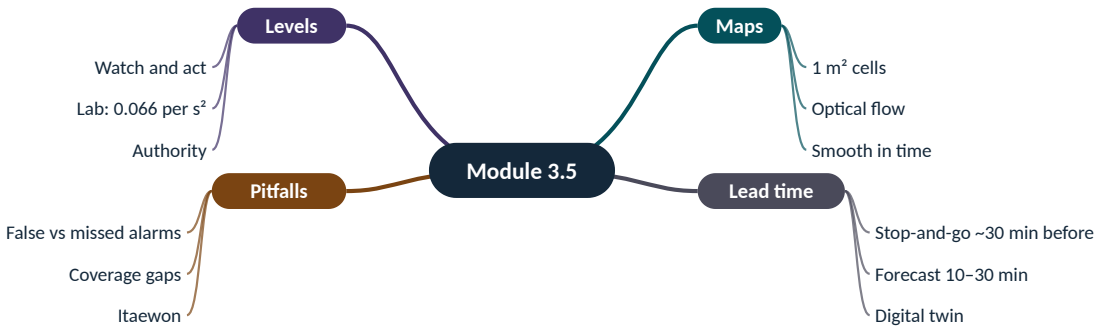
Control rooms turn camera and sensor data into density, flow and crowd-pressure maps, and act on thresholds before conditions become critical. Warning levels must be tested and tied to actions.

By the end you can

- compute density and crowd-pressure maps from video-derived data
- design warning and action levels tied to specific responses
- explain lead time and why early indicators matter
- describe how forecasts and digital twins support decisions

Overview

The best time to stop a crush is long before it happens. Control rooms at large events watch cameras and sensors and look for signs that density is rising towards critical levels. Video analysis can turn camera images into maps of density and local velocity. From these, crowd pressure, the local density times the variance of velocity, highlights places where a dense crowd has begun to move irregularly, which in the Jamarat data came before falls. A monitoring system needs levels: a watch level, at which staff look more closely, and an action level, at which a predefined response begins, such as stopping inflow, opening a relief route or making an announcement. Each level must be tied to an action and a person with the authority to take it. Lead time matters: a warning is only useful if there is time to act. Short-term forecasts, from models fed by live counts, and so-called digital twins of the venue can extend that lead time. In this module you will build a crowd-pressure map and set warning levels on it.



From video to maps

Split the scene into cells of about 1 m². In each cell and time window, estimate **density** (from a counting network or tracking) and **local velocity** (from optical flow or tracks). Then compute indicators per cell:

$$P_{\text{cell}} = \rho_{\text{cell}} \times \text{Var}(V_{\text{cell}}) \text{ over the last few seconds}$$

Smooth over time to avoid false alarms from single frames.

Levels tied to actions

A useful scheme has at least two levels: **watch** (look closely, prepare) and **act** (start the predefined response). In the lab, with a watch level of 0.01 and an action level of 0.02 per s^2, a synthetic dense zone near a bottleneck reaches 0.066, with 20 of 240 cells above the action level. Thresholds must be set from data and guidance for the site, tested in exercises, and reviewed after events. A level without an action, or an action without someone authorised to take it, is useless.

TRY Explore it

In the explorer, keep density at 7 per m^2 and lower the irregular motion. When does the alert switch off?

Lead time and forecasts

At Jamarat in 2006, stop-and-go began about 30 minutes, and turbulence about 10 minutes, before the first deaths (Helbing et al. 2007, as reported in the press). That is the window for action. Short-term forecasts extend it: a macroscopic or network model fed with live gate counts can predict density on a route 10 to 30 minutes ahead. A **digital twin** is such a model kept continuously in step with the site.

Pitfalls

False alarms erode trust; **missed alarms** cost lives. Camera coverage is never complete. Density estimates degrade in very dense crowds, at night and in rain. Operators can be overloaded with screens. At Itaewon, emergency calls warned of danger hours before the crush, but no effective action followed: monitoring only works when it is tied to response.

RESEARCH Open question

How to set and validate warning thresholds across different sites and cameras is an open research problem.

LAB Crowd-pressure map with alert levels

`labs/go3/m35_pressure.py`

Builds synthetic density and velocity fields over 50 frames, computes P = ρ·Var(V) per cell, and applies two levels.

```
max pressure 0.066 per s2 in cell (np.int64(14), np.int64(6))
cells above warning (0.01): 28, above action (0.02): 20 of 240
alert: ACT
```

Takeaways

- Turn video into density and velocity maps per cell, smoothed in time.
- Indicators such as crowd pressure cross predefined watch and action levels.
- Each level needs a defined action and someone with authority to take it.
- Forecasts and digital twins extend lead time; monitoring without response fails.

Check yourself

1. Crowd pressure is high when…
   (a) Density is high and motion is orderly · (b) Density is high and motion is irregular · (c) Density is low · (d) People stand still in a line
2. A warning level is useful only if…
   (a) It is very low · (b) It triggers a defined action by someone authorised · (c) It is secret · (d) It never fires
3. What extends lead time?
   (a) Bigger screens · (b) Short-term forecasts from live counts · (c) More barriers · (d) Later announcements
4. A digital twin is…
   (a) A copy of the stadium · (b) A model kept continuously in step with the live site · (c) A second control room · (d) A crowd counting camera

Answers: 1 (b) Both dense and irregular.; 2 (b) Tie levels to actions.; 3 (b) Forecasts warn earlier.; 4 (b) Model and site in step.

Words

- **Optical flow**: A computer-vision estimate of motion between video frames.
- **Watch level**: A threshold at which operators increase attention and prepare.
- **Action level**: A threshold that triggers a predefined response.
- **Lead time**: Time between a warning and the onset of danger.
- **Digital twin**: A model of a site kept continuously updated with live data.

Reading

- Helbing, Johansson and Al-Abideen (2007): Crowd pressure and warning indicators. https://arxiv.org/abs/physics/0701203
- Johansson et al. (2008): From crowd dynamics to crowd safety: Video-based monitoring at the Jamarat. https://arxiv.org/abs/0810.4590
- HSE: Managing crowds safely: Monitoring, communication and contingency plans. https://www.hse.gov.uk/pubns/books/hsg154.htm

3.6 Simulating a mass gathering

50 minutes · Methods

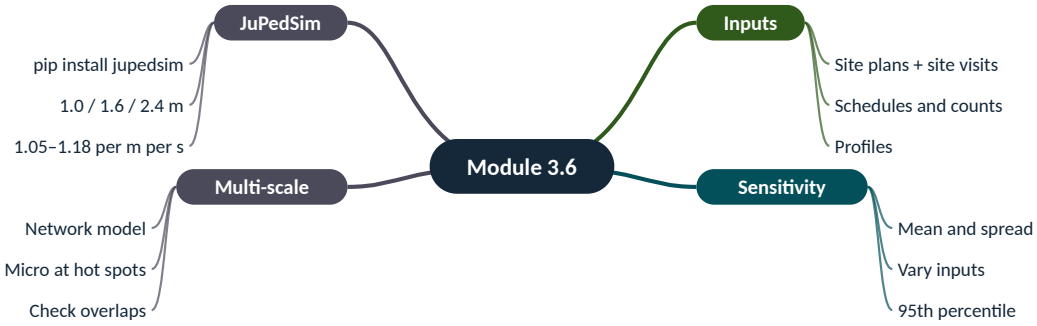
Build the geometry, set demand and schedules, choose and calibrate a model, run many seeds and scenarios, and analyse sensitivity. Open simulators such as JuPedSim make this reproducible.

By the end you can

- set up a simulation study from geometry to outputs
- run a scenario in an open-source simulator and repeat it with several seeds
- carry out a sensitivity analysis and report percentiles
- combine macroscopic and microscopic models across scales

Overview

A simulation of a mass gathering is a chain of choices, and each link can break. First the geometry: walkable areas, walls, gates, stairs and ramps, from site plans and checked on site. Then demand: how many people, arriving when, going where, often from schedules and counts. Then the model: macroscopic for the whole network, microscopic for hot spots, calibrated as in Go 2. Then the runs: stochastic models give different answers with different random seeds, so run each scenario several times. Then sensitivity: vary the uncertain inputs, such as pre-movement times, walking speeds and compliance with schedules, and see how much the outputs change. Finally the outputs that answer your question, such as risk exposure or time to clear, reported with their spread. Open-source simulators make this transparent and reproducible. In this module you run a gate scenario in JuPedSim, a simulator from Forschungszentrum Jülich, repeat it with several seeds and gate widths, and add a simple sensitivity analysis. JuPedSim reproduces bottleneck flows of about one point one people per metre per second here, consistent with experiments.



Geometry and demand

Build the geometry from site plans, then **check it on site**: temporary stalls, barriers and vehicles change effective widths. Demand comes from schedules, ticketing and counts: how many, when, from where to where, and with what profile (speeds, groups, mobility). For the Hajj, group schedules and routes are key inputs.

Running a scenario in JuPedSim

JuPedSim is an open-source simulator with a Python interface (`pip install jupedsim`). The lab builds a 20 m × 10 m forecourt emptying through a gate, places 150 people with the collision-free speed model, and repeats each gate width with 5 seeds. Results: 1.0 m gate about 143 s, 1.6 m about 80 s, 2.4 m about 57 s, with flows of about 1.05 to 1.18 people per metre per second, consistent with experiments.

TRY Run it yourself

Download the JuPedSim script, install JuPedSim, and add a 3.2 m gate. Is the flow per metre still about the same?

Seeds and sensitivity

Stochastic models differ between runs: report the **mean and spread** over seeds. Then vary uncertain inputs. In the explorer, ± 50% uncertainty in pre-movement and ± 20% in specific flow spread the egress time widely; the 95th percentile, not the median, is what safety depends on. Screen inputs one at a time first, then study the important ones jointly.

TRAP One run is an anecdote

A single stochastic run can be unusually good or bad. Never report one run as the result.

Across scales

No single model fits a whole mass gathering. A common approach couples a fast **network or macroscopic model** for the whole site and timetable with **microscopic models** for critical spots such as ramps, junctions and gates. The coupling passes flows and densities at the boundaries. Check that both models agree where they overlap.

LAB A gate scenario in JuPedSim

`labs/go3/m36_jupedsim.py`

Requires pip install jupedsim (version 1.x). Builds the geometry with shapely, runs the collision-free speed model, repeats each gate width with 5 seeds.

```
gate 1.0 m: evacuation 142.7 s (min 141.2, max 143.9), flow 1.05 per m per s
gate 1.6 m: evacuation 79.5 s (min 78.2, max 80.3), flow 1.18 per m per s
gate 2.4 m: evacuation 57.3 s (min 55.5, max 59.1), flow 1.09 per m per s
```

LAB Monte Carlo sensitivity of egress time

One-at-a-time and joint uncertainty. Which input widens the interval more?

```
pre +/-0%, flow +/-0%: median 6.8 min, 5-95% 6.8-6.8 min
...
```

Takeaways

- A simulation study chains geometry, demand, model, runs, sensitivity and outputs.

- Check geometry on site; derive demand from schedules and counts.
- Repeat stochastic runs and vary uncertain inputs; report percentiles.
- Couple macro models for the site with micro models for hot spots; share code and data.

Check yourself

1. Why run each scenario with several seeds?
   (a) To use more computer time · (b) Stochastic models vary between runs; report the spread · (c) Seeds make it faster · (d) It is required by JuPedSim
2. In the JuPedSim lab, a 1.6 m gate gave a flow of about…
   (a) 0.5 per m per s · (b) 1.2 per m per s · (c) 3 per m per s · (d) 8 per m per s
3. A common multi-scale approach couples…
   (a) Two identical micro models · (b) A network or macro model for the site with micro models at hot spots · (c) Weather and traffic models · (d) Only hand calculations
4. Which percentile matters most for safety?
   (a) 5th · (b) Median · (c) 95th or higher · (d) None

Answers: 1 (b) One run is an anecdote.; 2 (b) About 1.18, consistent with experiments.; 3 (b) Macro for scale, micro for detail.; 4 (c) The slow tail.

Words

- **Random seed**: A number that fixes the random sequence of a stochastic simulation run.
- **Sensitivity analysis**: Studying how outputs change with uncertain inputs.
- **JuPedSim**: Open-source pedestrian simulator from Forschungszentrum Jülich, with a Python interface.
- **Multi-scale coupling**: Linking models of different detail at their boundaries.
- **Reproducibility**: The ability of others to rerun a study and get the same results.

Reading

- JuPedSim: Documentation and tutorials for the open-source simulator. https://www.jupedsim.org/
- Vadere: Another open framework, with optimal steps and other models. https://www.vadere.org/
- Ronchi et al. (2013): V&V of evacuation models (NIST TN 1822): Testing and reporting simulation results. https://nvlpubs.nist.gov/nistpubs/TechnicalNotes/NIST.TN.1822.pdf

3.7 Interventions and their evaluation

45 minutes · Interventions

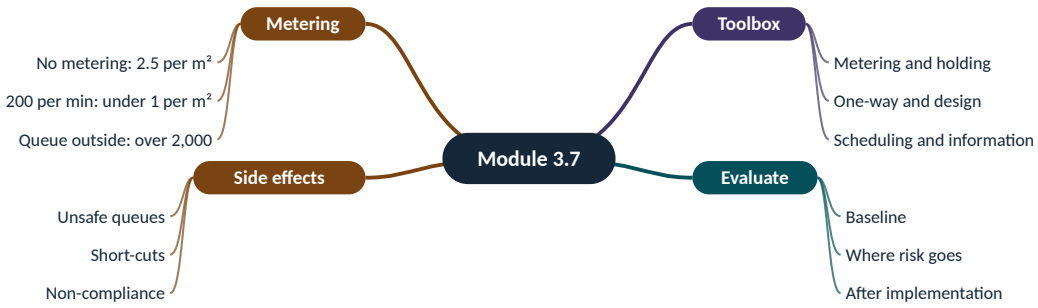
Metering, one-way systems, holding areas, scheduling and information can keep density down, but they move risk as well as reduce it. Evaluate them with simulation and data, including side effects.

By the end you can

- describe common crowd interventions and how they work
- show with a model how metering moves waiting from one place to another
- design an evaluation of an intervention with a fair comparison
- look for unintended effects

Overview

Once you understand where risk builds up, you can change the system. The most common tools are simple. Metering limits the rate at which people enter a space, so waiting happens in a safer place, often an open approach rather than an enclosed plaza. One-way systems separate opposing flows, as on the rebuilt Jamarat Bridge. Holding areas store people safely and release them in batches. Scheduling spreads arrivals over time. Information tells people where to go and when, and stewards make it real. But interventions do not make risk disappear; they move it. Metering creates a queue outside, which must itself be safe. A one-way system lengthens some routes. A holding area can become a crush if it is not released in time. That is why every intervention needs an evaluation: compare it fairly with a baseline, in simulation first and with data after, and look for unintended effects. In this module you will meter the gates in front of a plaza and see density inside fall while the queue outside grows.



The toolbox

Metering: admit people at a controlled rate. **One-way systems**: separate opposing flows. **Holding areas**: store and release in batches. **Scheduling**: spread demand over time. **Information and stewarding**: guide choices and enforce plans. **Design changes**: widen bottlenecks, remove obstacles, add routes. Most real plans combine several.

Metering moves the queue

A 1,200 m² plaza feeds a venue at 180 people per minute, with arrivals peaking at 320 per minute. Without metering the plaza peaks at about 2.5 per m². Metering the gates at 200 per minute keeps it under 1 per m², but a queue of over 2,000 builds outside (lab, 20 random seeds). Metering is only an improvement if the space outside is open, wide and managed.

TRY Explore it

Find the metering rate that keeps the plaza below 2 per m² with the smallest queue outside.

Evaluating fairly

Compare the intervention with a **baseline** under the same demand, using the same model and outputs, over several seeds and sensitivity cases. Report **where risk exposure goes**, not only whether it falls at the target. After implementation, monitor with data and compare with predictions. Real-world before–after comparisons must allow for other changes, such as different attendance.

Unintended effects

Look for: queues in unsafe places; longer routes that people refuse to follow (short-cuts through barriers); holding areas that fill faster than planned; counterflows created by people going back for friends; and information overload. People adapt to interventions, so simulate behavioural responses (for example non-compliance) as part of the sensitivity analysis.

RESEARCH Research angle

Few interventions at mass gatherings have been evaluated with before–after data. Designing such evaluations is a valuable research contribution.

LAB Gate metering with random arrivals

`labs/go3/m37_metering.py`

Sweeps the metering rate with 20 random seeds each and reports the plaza's peak density and the queue outside.

```
metering 400/min: plaza peak 2.47 per m2, queue outside 0
...
metering 200/min: plaza peak 0.86 per m2, queue outside 2,330
```

Takeaways

- Interventions: metering, one-way systems, holding areas, scheduling, information, design changes.
- Interventions move risk as well as reduce it; metering moves waiting upstream.
- Evaluate against a fair baseline and report where risk exposure goes.
- Include people's adaptation, such as non-compliance and short-cuts, in tests.

Check yourself

1. Metering the gates of a plaza mainly…
   (a) Removes all waiting · (b) Moves waiting from the plaza to the approach · (c) Raises density inside · (d) Has no

effect

2. A fair evaluation compares the intervention with…
   (a) A different event · (b) A baseline under the same demand and outputs · (c) Nothing · (d) Last year's weather
3. Which is an unintended effect of barriers?
   (a) Shorter routes · (b) People taking short-cuts around them · (c) Lower attendance · (d) Faster walking
4. In the lab, metering at 200 per minute gave a plaza peak of about…
   (a) 2.5 per m² · (b) under 1 per m² · (c) 5 per m² · (d) 0

Answers: 1 (b) The queue moves upstream.; 2 (b) Like with like.; 3 (b) People adapt.; 4 (b) About 0.86 per m².

Words

- **Metering**: Admitting people to a space at a controlled rate.
- **Holding area**: A space where people wait safely before release.
- **One-way system**: Routing that separates opposing flows.
- **Baseline**: The situation without the intervention, used for comparison.
- **Unintended effect**: A consequence of an intervention that was not its purpose.

Reading

- HSE: Managing crowds safely: Practical crowd control measures and planning. https://www.hse.gov.uk/pubns/books/hsg154.htm
- Johansson et al. (2008): Interventions at the Jamarat after 2006. https://arxiv.org/abs/0810.4590
- Still et al. (2020): Case studies of crowd safety interventions. https://e-space.mmu.ac.uk/625559/

3.8 The research frontier and writing it up

40 minutes · Research

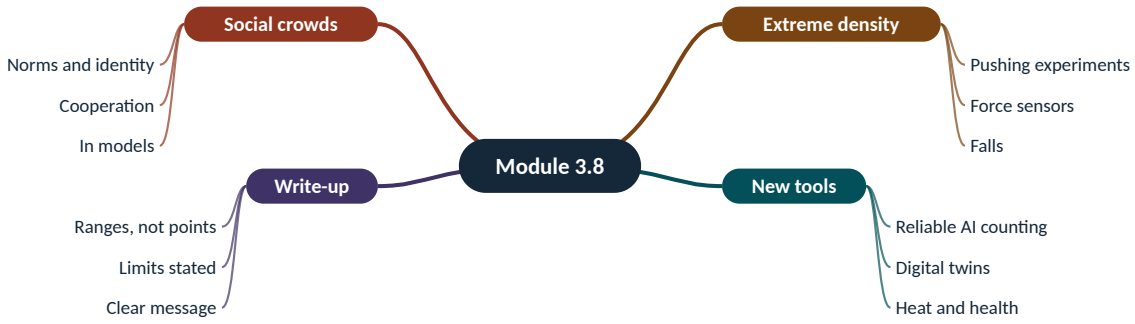
Open questions remain: pushing and high-density physics, the psychology of crowds, AI for monitoring, digital twins, and heat and health at mass gatherings. Report findings honestly, with uncertainty and limits.

By the end you can

- name open research questions in crowd dynamics and mass-gathering safety
- describe how physics and social psychology are being combined
- write a results section that states uncertainty and limits
- plan how to communicate findings to decision makers

Overview

Crowd science has come a long way since Fruin's levels of service, but many questions remain open. How exactly do pushing and contact forces spread through very dense crowds, and what densities and forces are truly critical? Laboratory experiments on pushing at entrances, and new sensors that measure body forces, are starting to answer this. How do social identity, culture and norms change movement, and how can models include them? Research increasingly combines pedestrian physics with social psychology. Can artificial intelligence count and track crowds reliably in real time, across cameras and cultures, while protecting privacy? Can digital twins of venues forecast danger early enough to act? How do heat and health interact with crowding at events such as the Hajj, where many participants are older? When you write up your own study, be honest about uncertainty and limits: what your data and model can and cannot show, and how sensitive your conclusions are. Decision makers need clear messages, ranges rather than single numbers, and a plain statement of what would change your conclusions.



Physics at extreme density

Most models were developed and calibrated at moderate densities. At extreme densities, contact forces, pushing and falls dominate. Laboratory experiments with pushing at entrances, and pressure sensors worn by participants, are

beginning to quantify how forces spread. Key questions: which densities and forces are critical, and how do falls start and spread?

Crowds are social

Pedestrian dynamics increasingly works with social psychology. Sieben, Schumann and Seyfried (2017) showed how social norms and identity change behaviour in experiments; Drury and colleagues showed cooperation in emergencies. Open questions: how to represent groups, culture and shared identity in models, and how to calibrate them.

RESEARCH Pick your front

Choose one front for your project, and say in your proposal which open question you address and how your data could answer it.

AI, digital twins and health

Deep learning has made crowd counting from images practical, but reliability across cameras, crowds and conditions, and privacy, remain open. Digital twins promise forecasts that extend lead time, but need validated models and good data. At the Hajj and other events in hot climates, heat and health interact with crowding and walking speeds, a frontier where crowd science meets mass-gathering medicine.

Writing it up

A strong results section states: the question; the data and model, with calibration and validation; the scenarios; results **with ranges** (seeds and sensitivity); what the model **cannot** represent and how that could bias results; and a clear message for decision makers. Avoid false precision ("egress time 6.83 min") and over-claiming ("proves the site is safe").

TRAP Over-claiming

A simulation cannot prove a site is safe. It can show that, under stated assumptions, a design keeps risk exposure below a threshold with a stated margin.

LAB Reporting results with ranges

Report medians and ranges, and how often one scenario beats the other, instead of a single number.

```
baseline : median 91,881, 90% range 70,439-120,260 person-minutes
metering : median 61,868, 90% range 35,101-84,479 person-minutes
reduction: median 31,076; in 90% of paired runs metering was better
```

Takeaways

- Open fronts: physics at extreme density, crowd psychology, AI monitoring, digital twins, heat and health.
- Pedestrian dynamics increasingly combines physics with social psychology.
- Report results with ranges, state model limits, avoid false precision.
- Give decision makers a clear message and say what would change it.

Check yourself

1. Why is extreme density a research frontier?
   (a) It never happens · (b) Most models were calibrated at moderate densities; forces and falls dominate at extreme ones · (c) Cameras cannot see it · (d) It is illegal to study
2. Which is good reporting practice?
   (a) One run, three decimals · (b) Medians and ranges over seeds and sensitivity cases · (c) Hiding limits · (d) Claiming proof of safety
3. Combining physics with social psychology means…
   (a) Ignoring groups · (b) Representing norms, identity and groups in crowd models · (c) Only using surveys · (d) Only using forces
4. A simulation can…
   (a) Prove a site is safe · (b) Show that, under stated assumptions, risk stays below a threshold with a margin · (c) Replace stewards · (d) Predict exact attendance

Answers: 1 (b) Physics changes at extreme density.; 2 (b) Ranges, not points.; 3 (b) Crowds are social.; 4 (b) Conditional, not absolute.

Words

- **False precision**: Reporting more digits than the uncertainty justifies.
- **Over-claiming**: Stating conclusions stronger than the evidence supports.
- **Social norm**: A shared expectation about how to behave.
- **Mass-gathering medicine**: The field studying health risks and care at mass gatherings.

Reading

- Sieben, Schumann and Seyfried (2017): Collective phenomena in crowds: Where pedestrian dynamics needs social psychology. https://doi.org/10.1371/journal.pone.0177328
- Haghani and Sarvi (2018): Crowd behaviour and motion: Review of empirical methods and open questions. https://doi.org/10.1016/j.trb.2017.06.017
- Memish et al. (2012): Lessons from the Hajj: Mass-gathering medicine. https://doi.org/10.1016/S1473-3099(11)70337-1

Questions for spaced practice

From the question bank: answer from memory, then check. Revisit after 1, 3, 7 and 14 days.

3.1 Framing a mass-gathering crowd study

1. **Recall:** Give the four parts of a PICO question.
   → Population, Intervention, Comparison, Outcome.
2. **Explain:** Why is risk exposure better than peak density as an outcome?
   → It combines how long and how many people spent time above a threshold, distinguishing a brief spike from a long dangerous period.
3. **Calculate:** 1,000 m² at 5 per m² for 12 minutes, all above a 4 per m² threshold. Exposure?
   → 5 × 1,000 × 12 = 60,000 person-minutes.
4. **Explain:** Why use a factorial design instead of changing one factor at a time?
   → It captures interactions, for example a layout that helps only when schedules slip.
5. **Apply:** What ethics measures would you plan for CCTV data at a pilgrimage?
   → Data minimisation, anonymisation (for example counting without identifying), secure storage, approval from authorities and ethics boards, and clear purposes.
6. **Explain:** Why are mass gatherings especially hard to study?
   → Experiments are impossible at full scale, data are scarce and sensitive, demand is synchronised by rituals or schedules, and crowds are diverse.

3.2 Lessons from crowd disasters

1. **Recall:** What happened at Hillsborough in 1989?
   → A gate was opened to relieve congestion outside the stadium and supporters flowed into central pens that were already full; 97 people died.
2. **Explain:** Why are crowd disasters called systemic failures?
   → Because many weaknesses in design, information and management interact and align; there is rarely a single cause or culprit.
3. **Recall:** What was distinctive about the Love Parade site?
   → A single ramp, reached through tunnels, was the only entrance and exit of the festival area, so inflow and outflow met there.
4. **Explain:** Why do death counts for Mina 2015 differ so much?
   → The official Saudi count (769) was never updated, while independent tallies from national governments and media exceeded 2,000; data access and transparency were limited.
5. **Apply:** How would you score Itaewon in DIM-ICE?
   → High risk in almost every cell: no designed routes or one-way systems, no information to the crowd, and no event management for ingress, circulation or egress.
6. **Explain:** What is hindsight bias and why does it matter?
   → Judging past decisions with knowledge decision makers did not have; it can lead to blaming individuals instead of fixing the information and systems they relied on.

3.3 The Hajj as a crowd system

1. **Recall:** Why is Hajj demand called synchronised?
   → Because the ritual sequence fixes the days and places of the rites, so very large numbers of pilgrims need the same places at nearly the same times.
2. **Recall:** How was the Jamarat Bridge redesigned after 2006?
   → As a multi-storey structure with several levels, wide ramps and one-way flows so that streams to and from the pillars do

not meet.

3. **Calculate:** 450,000 pilgrims, 6 equal slots of one hour, route capacity 60,000 per hour. Is it enough?
   → No: 75,000 per slot exceeds 60,000; use more slots or routes.
4. **Explain:** Why does non-compliance hurt scheduling so much?
   → Groups that leave early tend to choose the same popular times, concentrating extra demand in a few slots, which can push them far over capacity.
5. **Apply:** Name two research questions for Hajj crowd management.
   → How robust schedules are to non-compliance and delays; how real-time monitoring should adjust release times; how heat and age affect speeds and risk; how to use camera and phone data with privacy.
6. **Explain:** Why do pilgrim groups make scheduling possible?
   → Because groups have leaders and fixed organisers, authorities can assign and communicate time slots and routes to each group, unlike an anonymous crowd.

3.4 Data and sensing at mass gatherings

1. **Recall:** State Jacobs' method.
   → Divide the area into zones, multiply each zone's area by its estimated density, and add the zones.
2. **Calculate:** 10,000 m^2 at 2 per m^2 and 20,000 m^2 at 0.5 per m^2. Crowd size?
   → 20,000 + 10,000 = 30,000 people.
3. **Explain:** How does a density-map network count a crowd?
   → It predicts for each pixel the expected number of heads; summing the map gives the count, avoiding the need to detect every person.
4. **Explain:** Why report an interval instead of one number?
   → Densities, areas and model outputs are uncertain; an interval shows how far the true size could be from the estimate.
5. **Apply:** What biases affect phone-based crowd estimates?
   → Not everyone carries a phone or has it switched on; operators, settings and roaming differ; children and older people may be under-represented.
6. **Apply:** List three privacy safeguards for CCTV-based counting.
   → Count without identifying, aggregate to density per cell, keep data briefly and securely, and obtain approvals.

3.5 Real-time monitoring and early warning

1. **Recall:** Give the formula for crowd pressure in a cell.
   → P = local density × variance of local velocity over a short time window.
2. **Explain:** Why tie warning levels to predefined actions?
   → Because a warning without a planned response and an authorised decision maker leads to delay; predefined actions can start immediately.
3. **Explain:** What is lead time and why does it matter?
   → The time between a warning and danger; only warnings with enough lead time allow inflow to be stopped and crowds relieved.
4. **Calculate:** A cell has density 6 per m^2 and speed variance 0.004 m^2/s^2. Pressure?
   → 6 × 0.004 = 0.024 per s^2, above a 0.02 action level.
5. **Apply:** Name two pitfalls of camera-based monitoring.
   → Incomplete coverage, poor estimates in very dense crowds or bad light, false alarms eroding trust, operator overload, and warnings without response.
6. **Explain:** How can forecasts help a control room?
   → Models fed by live counts can predict density 10–30 minutes ahead, so staff can act before thresholds are reached.

3.6 Simulating a mass gathering

1. **Recall:** List the main steps of a crowd simulation study.
   → Geometry, demand, model choice and calibration, runs with several seeds, sensitivity analysis, outputs with spread, and documentation.
2. **Explain:** Why check geometry on site?
   → Temporary stalls, barriers, vehicles and closures change effective widths and routes, so plans alone can overestimate capacity.
3. **Recall:** What flows did the JuPedSim gate lab give?
   → About 1.05 to 1.18 people per metre per second, consistent with laboratory experiments.
4. **Explain:** What is the purpose of a sensitivity analysis?
   → To see how much outputs change with uncertain inputs, identify which inputs matter most, and report results with honest ranges.
5. **Apply:** Why couple a macroscopic model with microscopic ones for the Hajj?
   → The whole site and timetable need a fast network model, but ramps, junctions and gates need individual detail; coupling gives both.
6. **Apply:** What would you share to make a simulation study reproducible?
   → Scripts, geometry and parameter files, input data (or how to get it), simulator versions and random seeds.

3.7 Interventions and their evaluation

1. **Recall:** Name four types of crowd intervention.
   → Metering, one-way systems, holding areas, scheduling, information and stewarding, design changes.
2. **Explain:** Why do interventions move risk rather than delete it?
   → Because demand still has to go somewhere: metering creates queues upstream, holding areas store people, one-way systems lengthen routes; the new places must be safe.
3. **Calculate:** Arrivals peak at 320 per minute, venue takes 180, metering at 200. Roughly how fast does the outside queue grow at the peak?
   → About 320 − 200 = 120 people per minute.
4. **Explain:** What makes an evaluation fair?
   → Same demand, same model and outputs, several seeds and sensitivity cases, comparing intervention and baseline, and reporting exposure in all places.
5. **Apply:** Name two unintended effects to test for when adding barriers.
   → People taking short-cuts around them, new bottlenecks at barrier ends, longer routes that cause counterflows, and holding areas filling too fast.
6. **Apply:** When is metering a good idea?
   → When the space upstream is open, wide and managed, so waiting there is safer than in the enclosed space downstream.

3.8 The research frontier and writing it up

1. **Recall:** Name three open research fronts in crowd dynamics.
   → Pushing and forces at extreme density; social psychology and culture in models; AI for monitoring; digital twins; heat and health at mass gatherings.
2. **Explain:** Why is combining physics and social psychology important?
   → Crowd movement depends on norms, identity and groups as well as space and forces; models that ignore them can mispredict behaviour, especially in emergencies.
3. **Explain:** What is false precision and how do you avoid it?
   → Reporting more digits than the uncertainty supports; report rounded values with ranges from seeds and sensitivity.
4. **Apply:** What should a results section say about model limits?
   → What the model cannot represent (for example pushing, groups or information), how that could bias the results, and which assumptions would change the conclusions.

5. **Explain:** Why can a simulation not prove a site is safe?
   → Results depend on assumptions about demand, behaviour and parameters; a simulation can only show conditional performance under stated assumptions and margins.
6. **Apply:** How would you communicate a finding to an event authority?
   → A clear message, ranges rather than single numbers, the key assumptions, what would change the conclusion, and practical implications.